

$$5. (a) \quad \mathbb{Z}[\sqrt{5}] = \{a + b\sqrt{5} : a, b \in \mathbb{Z}\} \cong \frac{\mathbb{Z}[x]}{\langle x^2 - 5 \rangle}$$

$$\varphi : \mathbb{Z}[x] \longrightarrow \mathbb{Z}[\sqrt{5}]$$

$$f(x) \longmapsto f(\sqrt{5})$$

$$\left(\begin{array}{l} f(x) = g(x) \\ \underline{f(\sqrt{5})} = \underline{g(\sqrt{5})} \end{array} \right)^2$$

$$f(x) + g(x) \longmapsto f(\sqrt{5}) + g(\sqrt{5})$$

$$f(x)g(x) \longmapsto f(\sqrt{5})g(\sqrt{5})$$

$$a + b\sqrt{5} \longleftarrow \underline{a + bx = f(x)}$$

$$\frac{\mathbb{Z}[x]}{\ker \varphi} \cong \mathbb{Z}[\sqrt{5}] \text{ [by 1st Isomorphism thm]}$$

Claim: $\ker \varphi = \langle x^2 - 5 \rangle$

$$\langle x^2 - 5 \rangle \subseteq \ker \varphi$$

$$f(x) \in \langle x^2 - 5 \rangle$$

$$\Rightarrow f(x) = \underline{(x^2 - 5)} q(x)$$

$$\varphi(f(x)) = f(\sqrt{5}) = 0 \quad q(\sqrt{5}) = 0$$

$$f(x) \in \ker \varphi$$

$$\ker \varphi \subseteq \langle x^2 - 5 \rangle$$

$$f(x) \in \ker \varphi \Rightarrow f(\sqrt{5}) = 0$$

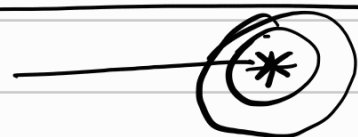
$$(x^2 - 5) \mid f(x) \quad [\because x^2 - 5 \text{ is minimal poly of } \sqrt{5}]$$

$$\Rightarrow f(x) \in \langle x^2 - 5 \rangle$$

$$\Rightarrow \ker \varphi \subseteq \langle x^2 - 5 \rangle$$

(a) is proved

(b) Every maximal ideal in commutative ring with identity $\Rightarrow$ prime ideal.



Solⁿ of part (b)

$$\Rightarrow \frac{\mathbb{Z}[x]}{\langle x^2 - 5 \rangle} \cong \mathbb{Z}[\sqrt{5}]$$

Correspondence Thm:

Every prime ideal of $R \xleftrightarrow{\text{1-1, onto}} \text{Every prime ideals of } \mathbb{Z}[x] \text{ containing } \langle x^2 - 5 \rangle$

in fact we can show

any prime ideal of R is of the form: $\frac{I}{\langle x^2 - 5 \rangle}$
 where I is prime ideal of $\mathbb{Z}[x]$ & $\langle x^2 - 5 \rangle \subseteq I$.

Let $\tilde{I}$ be that prime ideal

$$\Rightarrow 6 \in \tilde{I} \trianglelefteq R \cong \frac{\mathbb{Z}[n]}{\langle n^2-5 \rangle}$$

Now, $6 \in \tilde{I} \cong \frac{\tilde{I}}{\langle n^2-5 \rangle}$

$\downarrow$

$6 + \langle n^2-5 \rangle \in \frac{\tilde{I}}{\langle n^2-5 \rangle}$

$$\Rightarrow 6 \in \tilde{I}$$

$$\Rightarrow \langle n^2-5 \rangle \subseteq \tilde{I}$$

$$\Rightarrow \langle 6, n^2-5 \rangle \subseteq \tilde{I}$$

$$\varphi: \mathbb{Z}[n] \rightarrow \mathbb{Z}_2$$

$$\frac{\langle n^2-5 \rangle}{\langle n^2-5 \rangle} \rightarrow \leftarrow 6$$

$$2, 3 \in \tilde{I} \Rightarrow \text{either } 2 \in \tilde{I} \text{ or } 3 \in \tilde{I}$$

$$I \subseteq \frac{\langle 2, n^2-5 \rangle}{\langle n^2-5 \rangle} \subseteq \tilde{I}$$

$$\text{or } I \subseteq \frac{\langle 3, n^2-5 \rangle}{\langle n^2-5 \rangle} \subseteq \tilde{I}$$

Claim: either $I = \frac{\langle 2, n^2-5 \rangle}{\langle n^2-5 \rangle}$ ✓

or $I = \frac{\langle 3, n^2-5 \rangle}{\langle n^2-5 \rangle}$ ✓

$I = R$ ✓

$$\frac{\mathbb{Z}[n]}{\langle 2, n^2-5 \rangle} \cong \frac{\frac{\mathbb{Z}[n]}{\langle 2 \rangle}}{\langle 2, n^2-5 \rangle / \langle 2 \rangle} \cong \frac{\mathbb{Z}_2[n]}{\langle n^2-5 \rangle} \cong \mathbb{Z}_2[\sqrt{5}]$$

$$\frac{\mathbb{Z}[\alpha]}{\langle 3, \alpha^2 - 5 \rangle} \cong \mathbb{Z}_3[\sqrt{5}] \text{ a field.}$$
